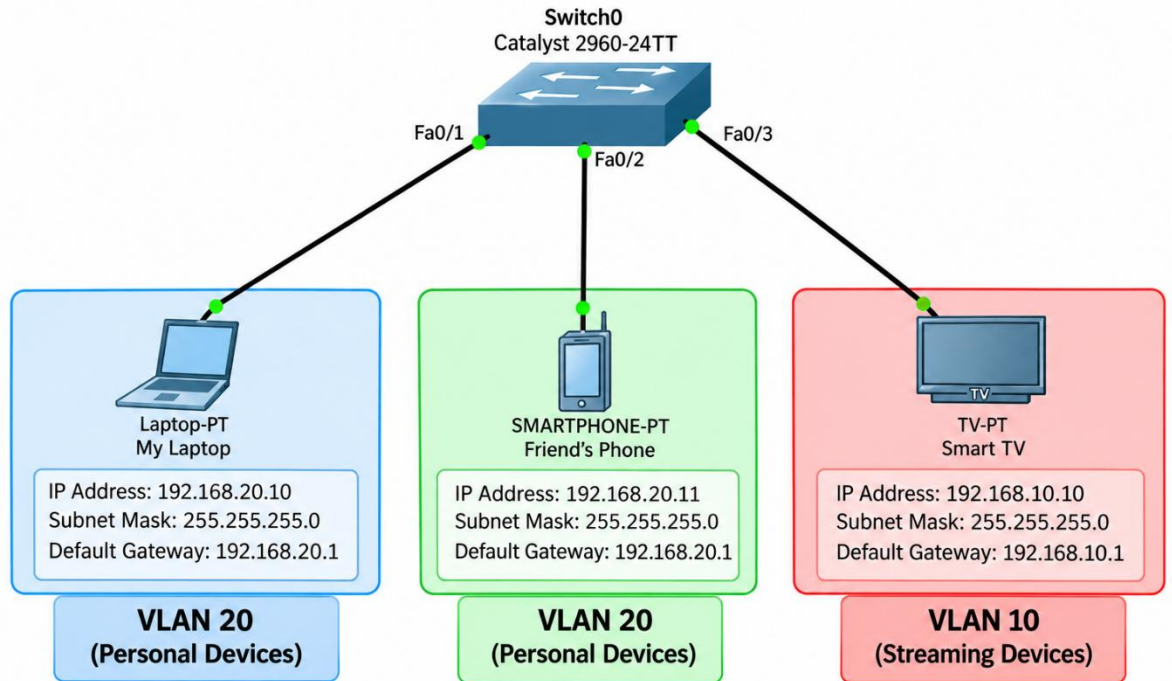
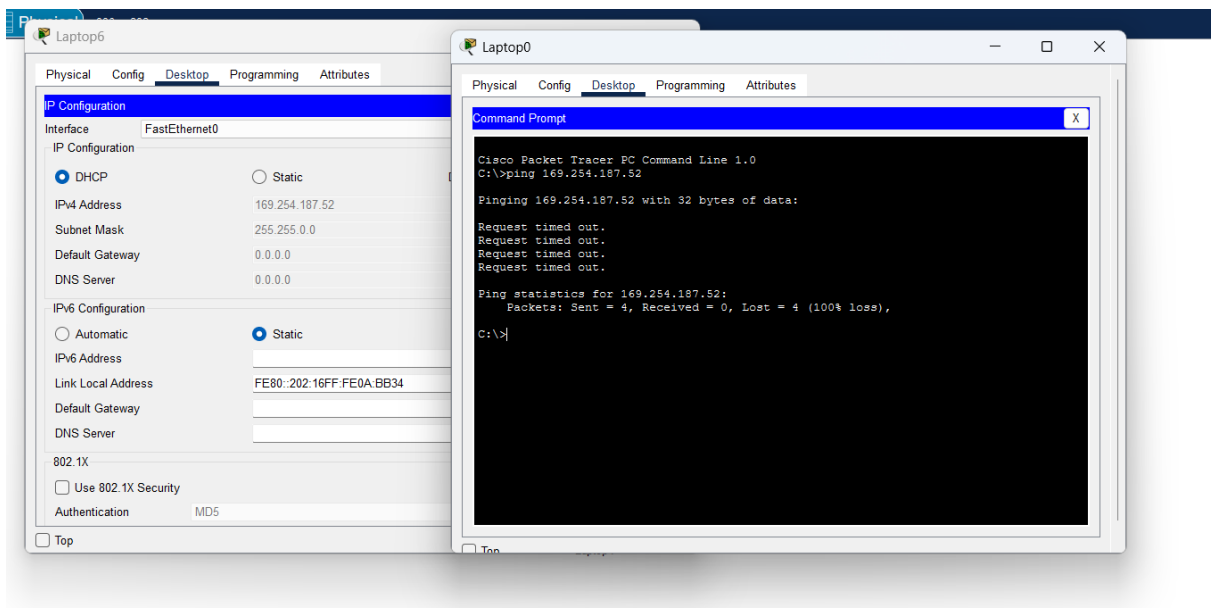


Switching and Vlans

1. Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).



2. Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other.



VLAN Configuration and Verification

I used Cisco Packet Tracer to configure a switch with two VLANs: **VLAN 10** and **VLAN 20**. I created three PCs in VLAN 10 and two PCs in VLAN 20. The switch ports connected to the PCs were configured as access ports and assigned to their respective VLANs.

Switch configuration:

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# vlan 10
```

```
Switch(config-vlan)# name VLAN10
```

```
Switch(config-vlan)# exit
```

```
Switch(config)# vlan 20
```

```
Switch(config-vlan)# name VLAN20
```

```
Switch(config-vlan)# exit
```

```
Switch(config)# interface range fa0/1-3
```

```
Switch(config-if-range)# switchport mode access
```

```
Switch(config-if-range)# switchport access vlan 10
```

```
Switch(config-if-range)# exit
```

```
Switch(config)# interface range fa0/4-5
```

```
Switch(config-if-range)# switchport mode access
```

```
Switch(config-if-range)# switchport access vlan 20
```

```
Switch(config-if-range)# exit
```

PC Assignment

PC	VLAN	Example IP
PC1	VLAN 10	192.168.10.2
PC2	VLAN 10	192.168.10.3
PC3	VLAN 10	192.168.10.4
PC4	VLAN 20	192.168.20.2
PC5	VLAN 20	192.168.20.3

I assigned IP addresses manually so that the PCs in each VLAN are in different networks.

Verification

I tested connectivity using the ping command. PCs belonging to the **same VLAN** were able to communicate with each other. However, when I tried to ping a PC from **VLAN 10** to a PC in **VLAN 20**, the ping failed/time-out occurred.

3. In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

I used Cisco Packet Tracer and moved a PC from VLAN 10 to VLAN 20.

Before:

```
Switch(config)# interface fa0/1
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 10
```

The PC was in **VLAN 10** and could communicate with other VLAN 10 PCs.

After:

```
Switch(config)# interface fa0/1
```

```
Switch(config-if)# switchport access vlan 20
```

The PC was moved to **VLAN 20**. I verified using:

```
Switch# show vlan brief
```

The PC could communicate with VLAN 20 PCs but not directly with VLAN 10 PCs.

Result: The PC was successfully moved from VLAN 10 to VLAN 20 and the connectivity change was verified.

4. Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3).

I used **Cisco Packet Tracer** and assigned ports **Fa0/1 to Fa0/10** to VLAN 30 using the interface range command.

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# vlan 30
```

```
Switch(config-vlan)# name VLAN30
```

```
Switch(config-vlan)# exit
```

```
Switch(config)# interface range fa0/1-10
```

```
Switch(config-if-range)# switchport mode access
```

```
Switch(config-if-range)# switchport access vlan 30
```

```
Switch(config-if-range)# exit
```